Guidelines:

In every subject:

a) In Section-A, 35 Questions will be given. Answer all 35 questions from Section-A

b) In Section-B, 15 questions will be given. Out of which answer 10 question only

BOTANY

SECTION - A

1. Read the following statements and find out the incorrect statements with reference to agarose gel electrophoresis.

   i) Now – a – days the most commonly used matrix is agarose which is a synthetic polymer extracted from sea weeds.

   ii) Larger the fragment size, the farther it moves.

   iii) We cannot see pure DNA in UV light without staining

   iv) DNA fragments separate (resolve) according to their size through sieving effect provided by the agarose gel.

   (1) i, ii and iii (2) ii, iii and iv

   (3) iii and iv (4) i and ii

2. The green revolution succeeded in tripling the food supply but yet it was not enough to feed the growing human population. Increased yields have been due to the use of :

   a. Improved crop varieties

   b. Better management parctices

   c. Agrochemicals (fertilizers and pesticides)

   Choose the correct option

   (1) a and b (2) b and c

   (3) a and c (4) a, b and c

3. The term used to refer to the use of bio-resources by multinational companies and other organizations without proper authorization from the countries and people concerned without compensatory payment is

   (1) Bioprospecting

   (2) Biopiracy

   (3) Biofortification

   (4) Bioinformatics

4. How many documented varieties of Basmati rice are grown in India?

   (1) 2,00,000 (2) 50,000

   (3) 1,000 (4) 27

5. *Bacillus thruringiensis* has been used to kill certain insects such as

   a. Tobacco budworm

   b. Armyworm

   c. Beetles

   d. Flies

   e. Mosquitoes

   (1) 4 (2) 3

   (3) 2 (4) 5

6. Match the organism with its use in biotechnology.

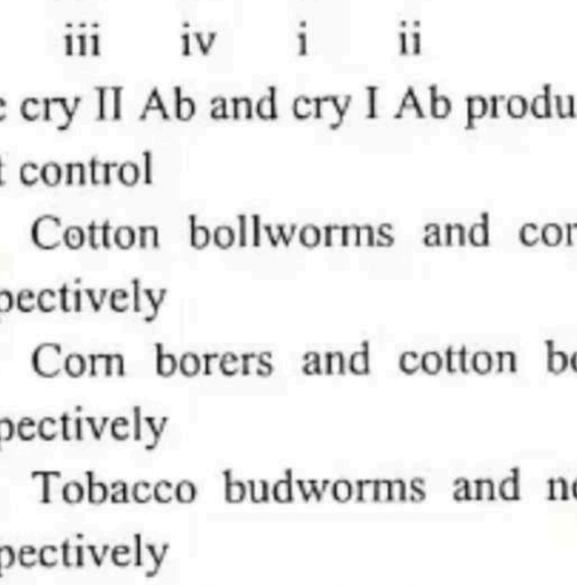
Column – I		Column – II	
a)	*Bacillus thuringiensis*	i)	Cloning vector
b)	*Thermus aqaticus*	ii)	Related to construction of first rDNA molecules
c)	Modified Ti plasmid of *Agrobacterium tumefaciens*	iii)	DNA polymerase (Taq)
d)	*Salmonella typhimurium*	iv)	Cry proteins

Select the correct option from the following :

	a	b	c	d
(1)	ii	iv	iii	i
(2)	iv	iii	i	ii
(3)	iii	ii	iv	i
(4)	iii	iv	i	ii

7. The cry II Ab and cry I Ab produce toxins that control
(1) Cotton bollworms and corn borers respectively
(2) Corn borers and cotton bollworms respectively
(3) Tobacco budworms and nematodes respectively
(4) Nematodes and tobacco budworms respectively

8. Recognise the figure and find out the correct matching.



(1) a – BamH I, b – Pvu I, c – Pst I, d – Sal I
(2) a – Pvu I, b – BamH I, c – Pst I, d – Sal I
(3) a – BamH I, b – Pvu I, c – Sal I, d – Pst I
(4) a – Pvu I, b – BamH I, c – Sal I, d – Pst I

9. Assertion : Traditional hybridisation procedures used in plant and animal breeding, very often lead to inclusion and multiplication of undesirable genes along with the desired genes.
Reason : The techiniques of genetic engineering cannot overcome this limitation.
(1) Both assertion and reason are true and the reason is the correct explanation of the assertion
(2) Both assertion and reason are true, but reason is not the correct explanation of the assertion
(3) Assertion is true, but reason is false
(4) Both assertion and reason are false

10. The insecticidal protein in Bt cotton binds to _____ of the insects
(1) Foregut epithelial cells
(2) Midgut mesothelial cells
(3) Midgut epithelial cells
(4) Hindgut mesothelial cells

11. Which of the following are considered as useful selectable markers for *E.coli*?
a. Ampicillin resistance gene
b. Chloramphenicol resistance gene
c. Tetracycline resistance gene
d. Kanamycin resistance gene
Choose the correct option
(1) a only (2) a and b
(3) a, b and c (4) a, b, c and d

12. Purified DNA ultimately precipitate out after the addition of chilled ethanol. This DNA that separates out can be removed by
(1) Electorphoresis
(2) Downstream processing
(3) PCR
(4) Spooling

13. A stirred-tank bioreactor is usually cylindrical or with a curved base to
(1) Facilitate the mixing of the reactor contents
(2) Control the temperatrue of the reactor
(3) Control the pH of the reactor
(4) Control oxygen entry into bioreactor

14. Which of the following is a biotechnological process?
(a) IVF leading to 'test tube' baby
(b) Developing a DNA vaccine
(c) Synthesising a gene and using it
(1) Only a (2) Only a, b
(3) Only b, c (4) a, b, c

15. Stickiness of the ends of the DNA facilitates the action of
(1) DNA ligase and these ends joined together laterally
(2) DNA ligase and these ends joined together end-to-end
(3) Taq polymerase
(4) Restriction enzyme

16. Match the following.

Column – I		Column – II	
a)	Thermostable DNA polymerase	1)	DNA ligase
b)	Joining the DNA molecules	2)	Bioreactors
c)	Large scale production of products	3)	Heterologous host
d)	Recombinant protein	4)	Taq polymerase

	a	b	c	d
(1)	4	1	2	3
(2)	3	2	1	4
(3)	2	4	3	1
(4)	1	3	4	2

17. There are three basic steps in genetically modifying an organism. Arrange these steps in correct sequence.
a. Introduction of the identified DNA into the host.
b. Maintenance of introduced DNA in the host and transfer of the DNA to its progeny
c. Identification of DNA with desirable genes
(1) $a \rightarrow b \rightarrow c$ (2) $b \rightarrow c \rightarrow a$
(3) $c \rightarrow b \rightarrow a$ (4) $c \rightarrow a \rightarrow b$

18. In restriction enzymes (like EcoRI the Roman numbers following the names indicate the:
(1) Order in which the enzyme were discovered from that strain of bacteria.
(2) Order in which the enzyme were isolated from that strain of bacteria.
(3) Genus of the prokaryotic cell or bacteria
(4) Strain of the bacteria

19. Asssertion : Besides contributing in growth, cell division also occurs in certain animal tissues to replace lost cells.
Reason : Growth can be taken as a defining property of living organisms.
(1) Both assertion and reason are true and the reason is the correct explanation of the assertion
(2) Both assertion and reason are true, but reason is not the correct explanation of the assertion
(3) Assertion is true, but reason is false
(4) Both assertion and reason are false

20. Which of the following taxonomical aid serve as a quick referral system in taxonomical studies?
(1) Key
(2) Botanical gardens
(3) Herbaria
(4) Museum

21. Fishes, amphibians, reptiles, birds and mammals together placed in the same category, called
(1) Family (2) Phylum
(3) Order (4) Class

22. Which of the following organisms are multiplied by fragmentation?
1. Fungi
2. Algae
3. Protonema of mosses
(1) 1 and 2 (2) 1, 2 and 3
(3) 2 and 3 (4) 1 and 3

23. I. Unicellular, colonial, filamentous, marine or terrestrial forms
II. The colonies are surrounded by a gelatinous sheath
III. Some can fix atmospheric nitrogen in specialised cells called heterocysts
IV. They often form blooms in water bodies
These above characters are seen in
(1) Archaebacteria
(2) Cyanobacteria
(3) Chryosophytes
(4) Dinoflagellates

24. Read the following statements and select the incorrect statement.
(1) Though the bacterial structrue is very simple, they are very complex in behaviour
(2) Chemosynthetic autotrophic bacteria play a great role in recycling nutrients like nitrogen, phosphorous, iron and sulphur
(3) Majority of the heterotrophic bacteria are important decomposers
(4) Autotrophic bacteria are only chemoautotrophs

25. Read the following statements and select the incorrect statement.
(1) Dinoflagellates are mostly marine while majority of euglenoids are found in running fresh water organisms
(2) The pigments of euglenoid are identical to those present in higher plants
(3) Amoeboid protozoans live in fresh water, sea water or moist soil.
(4) Marine amoeboids have silica shells on their surface.

26. Match the following w.r.t five kingdom system of classification

Column – I		Column – II	
a)	*Chlamydomonas*	i)	Plantae
b)	*Amoeba*	ii)	Animalia
c)	*Nostoc*	iii)	Protista
d)	Venus fly trap	iv)	Monera

	a	b	c	d
(1)	i	ii	iii	iv
(2)	iii	ii	iv	i
(3)	ii	iii	iv	i
(4)	i	ii	iv	i

27. Fill in the blanks.
The form and size of algae is highly variable, ranging from colonial forms like …….(a)….. and the filamentous forms like ……(b)…. .
(1) a – *Volvox*; b – *Ulothrix* and *Spirogyra*
(2) a – *Ulothrix* and *Spirogyra*; b – *Volvox*
(3) a – *Ulothrix*; b – *Volvox* and *Spirogyra*
(4) a – *Volvox* and *Spirogyra*; b – *Spirogyra*

28. The diploid sporophyte is represented by a dominant, independent, photosynthetic, vascular plant body. It alternates with multicellular, saprophytic/autotrophic, independent but short-lived haploid gametophyte. This type of pattern is exhibited by
(1) Bryophytes (*Sphagnum* and *Polytrichum*)
(2) Pteridophytes (Seleginella and *Lycopodium*)
(3) Most of the algal genera (*Fucus, Chara* and *Polysiphonia*)
(4) Seed plants (gymnosperms and angiosperms)

29. Read the following statements:
a) This is an exceptionally large group of plants occurring in wide range of habitats.
b) They provide us with food, fodder, fuel, medicines and several other commercially important products.
c) · They range in size from tiny, almost microscopic to tall trees over 100 metres.
Here, we are talking about
(1) Bryophytes (2) Pteridophytes
(3) Gymnosperms (4) Angiosperms

30. In *Bryophyllum*, the buds that arise from the notches of margins of leaves are called
(1) Apical buds
(2) Axillary buds
(3) Adventitious buds
(4) Terminal buds

31. Consider the following statements regarding gymnosperms and choose the correct option.
I. In gymnosperms, the male and female gametophytes have an independent existence
II. The nulticellular female gametophyte retained within the megasporangium
III. All gymnosperms are heterosporous.
(1) I and II are true but III is false
(2) I and III are true but II is false
(3) II and III are false but I is true
(4) II and III are true but I is false

32. Which of the following is incorrectly matched?
(1) *Porphyra* – Floridean starch
(2) *Volvox* – Starch
(3) *Ectocarpus* – Fucoxanthin
(4) *Ulothirx* – Mannitol

33. Which is not an example of vegetative propagule in angiosperms?
(1) Zygospores in algae
(2) Eyes of Potato
(3) Rhizome of Ginger
(4) Bulbil of *Agave*

34. Statement I : In Rhodophyceae, food is stored as mannitol and laminarin
Statement II : Ovules of gymnosperms are not enclosed by ovary wall.
(1) Both statement I and statement II are true
(2) Both statement I and statement II are false
(3) Statement I is true but statement II is false
(4) Statement I is false but statement II is true

35. In pheophyceae, the spores (zoospores) are
(1) Pyriform and bear 2 flagella (one longitudinal and other transverse)
(2) Pear – shaped and bear 2 flagella that are laterally attached
(3) Pyriform and bear 2-8, and equal and apical flagella
(4) Pear-shaped and bear 2-8, and unequal and apical flagella

SECTION – B

36. EcoRI cut the DNA between the bases
(1) G and T only (2) G and C only
(3) A and T only (4) G and A only

37. Which of the following contain cry gene?
(1) Bt Cotton (2) Mango
(3) Golden Rice (4) Solan gola

38. Assertion : Water hyacinth spreads all over the water body in a short period of time and it is very difficult to get rid of them.
Reason : Water hyacinth propagate vegetatively at a phenomenal rate.
(1) Both assertion and reason are true and the reason is the correct explanation of the assertion
(2) Both assertion and reason are true, but reason is not the correct explanation of the assertion
(3) Assertion is true, but reason is false
(4) Both assertion and reason are false

39. After fertilization, resulting in zygote, ovules and ovary develop into another specialised structure. Select the correct matching w.r.t post fertilisation events

(1) Zygote → Seed

(2) Ovary → Fruit

(3) Ovule → Embryo

(4) Sepals, petals and stamens → Remains attached

40. Fill in the blanks :

1. The …a… reproduction is the common mode of reproduction in organisms that have a relatively simple organization like algae and fungi and they shift to …b…. method of reproduction just before the onset of adverse conditions.

2. Asexual (vegetative) as well as sexual modes of reproduction are exhibited by the ….c….

3. Only sexual mode of reproduction is present in most of the ….d…

(1) a – sexual, b – asexual, c – higher plants, d – animals

(2) a – sexual, b – asexual, c – animals, d – higher plants

(3) a – asexual, b – sexual, c – higher plants, d – animals

(4) a – asexual, b – sexual, c – animals, d – higher plants

41. Consider the following statements and choose the correct option.

I. In bryophytes, sporophyte is having independent existence

II. Female gametophyte is retained within the megasporagium in angiosperms

III. All angiosperms are heterosporous

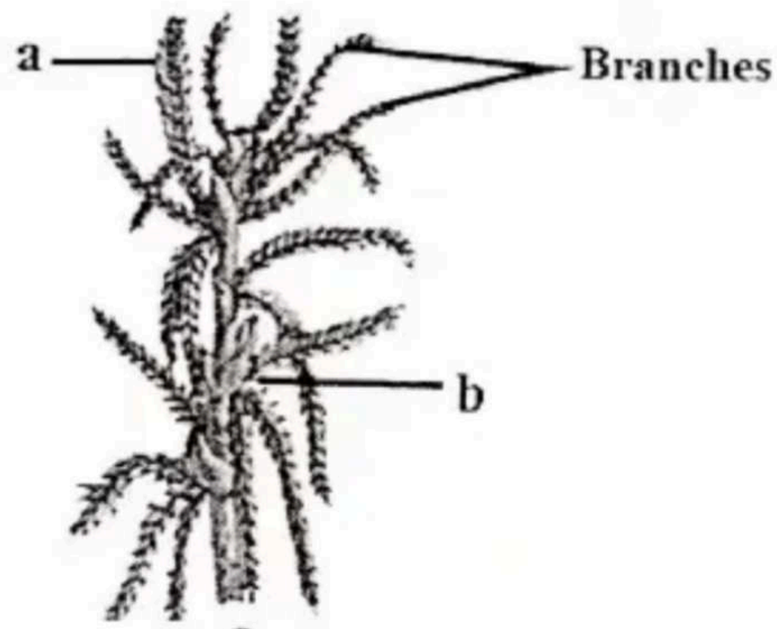
(1) I and II are true but III is false

(2) I and III are true but II is false

(3) II and III are false but I is true

(4) II and III are true but I is false

42. Recognise the figure and find out the correct matching :



(1) a – Antheridial branch, b – Archegonial branch, c – *Sphagnum* gametophyte

(2) a – Antheridial branch, b – Archegonial branch, c – *Sphagnum* sporophyte

(3) a – Archegonial branch, b – Antheridial branch, c – *Sphagnum* sporophyte

(4) b – Antheridial branch, a – Archegonial branch, c – *Funeria* gametophyte

43. Which of the following is the characteristic featrue of gymnosperms?

(1) Winged seeds

(2) Living fossils

(3) Naked seeds

(4) Multiciliated male gametes

44. Match the following.

Column – I		Column – II	
a)	Phycomycetes	i)	Sac fungi
b)	Ascomycetes	ii)	Imperfect fungi
c)	Basidiomycetes	iii)	Algal fungi
d)	Deuteromycetes	iv)	Club fungi/bracket

	a	b	c	d
(1)	i	iii	iv	ii
(2)	iii	i	iv	ii
(3)	i	iii	ii	iv
(4)	iii	i	ii	iv

45. Select the wrong statement

(1) In Diatoms, the walls are embedded with silica and thus, the walls are destructible.

(2) Red tides are due to red dinoflagellates

(3) Euglenoids are facultative heterotrophs.

(4) The spores of slime mould posses true walls

46. Statement I : In 5 kingdom system of classification unicellular eukaryotes were placed in a separate kingdom i.e. Fungi.

Statement II : Five kingdom system of classification was given by Carolus Linnaeus

(1) Both statement I and statement II are true

(2) Both statement I and statement II are false

(3) Statement I is true but statement II is false

(4) Statement I is false but statement II is true

47. *Rhizopus* belongs to class

(1) Oomycetes

(2) Phycomycetes

(3) Ascomycetes

(4) Basidiomycetes

48. What is cause of loose smut disease?

(1) *Puccinia* (2) *Ustilago*

(3) *Albugo* (4) *Mucor*

49. Select the correctly written scientific name of Mango which was first described by Carolus Linnaeus.

(1) *Mangifera indica* Linn.

(2) *Mangifera indica*

(3) *Mangifera Indica*

(4) *Mangifera indica* Car. Linn

50. Match the items given in Column I with those in Column II and select the correct option given below

Column – I		Column – II	
a)	Herbarium	i)	It is a place having a collection of preserved plants and animals
b)	Key	ii)	A list enumerates methodically all the species found in an area with brief description aiding identification.
c)	Museum	iii)	Is a place where dried and pressed plant specimens mounted on sheets are kept.
d)	Catalogue	iv)	A booklet containing a list of characters and their alternates which are helpful in identification of various taxa.

	a	b	c	d
(1)	i	iv	iii	ii
(2)	iii	ii	i	iv
(3)	ii	iv	iii	i
(4)	iii	iv	i	ii

ZOOLOGY

SECTION – A

51. How many teeth appear twice during the life span of human

(1) 20 (2) 12

(3) 24 (4) 32

52. Given below are two statements: one is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion (A) : Enhanced CO_2 concentration in the atmosphere is one of the major effects which are caused due to deforestation

Reason (R): Trees that could hold a lot of carbon in their biomass are lost with deforestation

In the light of the above statements, choose the correct answer from the options given below

(1) A is correct but R is not correct

(2) A is not correct but R is correct

(3) Both A and R are correct and R is the correct explanation of A

(4) Both A and R are correct but R is not the correct explanation of A

53. The following features are seen in which of the below options?
a) Exoskeleton of chitin
b) Malpighian tubules as excretory organ
c) Tracheal system for respiration
d) Three pair of legs in thoracic region
(1) Limulus (2) Prawn
(3) Scorpion (4) Bombyx

54. Intrinsic factor essential for absorption of vitamin B_{12} is secreted by
(1) Parietal cells (2) Chief cells
(3) G – Cells (4) Neck cells

55. Given below are two statements
Statement I: The food thoroughly mixed with the acidic gastric juice in the stomach is called chyme
Statement II: Mucus and bicarbonates of gastric juice protect the mucosal epithelium from excoriation by the highly concentrated HCl.
In the light of the above statements, choose the most appropriate answer from the options given below
(1) S-I and S-II are true
(2) S-I and S-II are false .
(3) S-I is true, but S-II is false
(4) S-I is false, but S-II is true

56. Study the following statements.
a) Specialised connective tissues include cartilage, bone and blood.
b) In all connective tissues except blood, the cells secrete fibres and these cells also secrete modified polysaccharides
c) Compound epithelium has a limited role in secretion and absorption.
d) Salivary glands are the unicellular glands.
In the above,
(1) All are true except d
(2) All are false except c
(3) All are true except a
(4) All are true except b

57. Which of the following are activated by enzymes in alkaline medium
a) pepsinogen b) trypsinogen
c) chymosin
d) chymotrypsinogen e) prorennin
(1) a, b, d (2) b, d
(3) a, b, d, e (4) b, c, d

58. Choose the incorrect statement
(1) Both mechanical and chemical process of digestion are initiated in the oral cavity
(2) About 30% of starch is hydrolysed into maltose in the oral cavity
(3) Mucus of the saliva helps in the formation of bolus in the oral cavity
(4) Pancreatic amylase acts on disaccharides to convert them into monosaccharides

59. Fatty acids and glycerol are incorporated into small droplets called micelles inside the
(1) intestinal mucosal cells
(2) blood capillaries
(3) intestinal lumen
(4) Lacteals

60. Protein deficiency unaccompanied by calorie deficiency is
(1) Marasmus (2) Constipation
(3) Kwashiorkar (4) Rickets

61. Given below are two statements: one is labelled as Assertion (A) and other is labelled as Reason (R).
Assertion (A): In spite of having more than 70 per cent of the world livestock population, the contribution of India and china to the world farm produce is only 25 per cent.
Reason (R): The productivity per unit of livestock in these countries is very high
In the light of the above statements, choose the correct answer from the options given below
(1) A is correct but R is not correct
(2) A is not correct but R is correct
(3) Both A and R are correct and R is the correct explanation of A
(4) Both A and R are correct but R is not the correct explanation of A

62. Which of the following is the most important factor that would lead to increased milk yield in cattle?
(1) Quality of breeds
(2) Resistance to diseases
(3) Stringent cleanliness and hygiene
(4) Provision of ideal environmental conditions to cattle

63. Among the following the number of marine water fishes is:
Catla, Rohu, Common carp, Hilsa, Sardines, Mackerel, Pomfrets
(1) 4 (2) 3
(3) 2 (4) 5

64. The best breeding method for animals that are below average in productivity in milk production would be
(1) Inbreeding
(2) Out-crossing
(3) Cross-breeding
(4) Interspecific hybridization

65. Human protein i.e. $\alpha-1$-antitrypsin obtained from transgenic animal is used to treat
(1) Rheumatoid arthritis
(2) Myasthenia gravis
(3) Glomerulonephritis
(4) Emphysema

66. Of the following which one is NOT the characteristic of triploblastic animals?
(1) All of them have blood vascular system
(2) They may be coelomates, pseudocoelomates or acoelomates
(3) They show bilateral symmetry
(4) They have digestive system

67. In *Scypha,* fertilization takes place in
(1) Mesoglea (2) Mesenchyme
(3) Mesohyal (4) Spongocoel

68. Metamerism in the body first evolved in which of the following?
(1) Platyhelminthes (2) Aschelminthes
(3) Annelida (4) Arthropoda

69. Notochord is persistent throughout life in
(1) *Ascidia* (2) Rohu
(3) *Ichthyophis* (4) *Branchiostoma.*

70. Common characteristic of all vertebrates without exception is–
(1) Presence of exoskeleton
(2) Presence of chambered heart
(3) Presence of renal portal system
(4) Internal fertilisation

71. Identify the incorrect one about electrostatic precipitator
(1) it has electrode wires that produce a corona releasing electrons
(2) electrons attach to dust particles and give them a net negative charge
(3) collecting plates are grounded to attract the charged dust particles
(4) the velocity of air passing between the collecting plates must be very high

72. Catalytic converters reduce the emission of poisonous gases from automobiles, by converting
(1) CO_2 into CO
(2) Hydrocarbons into CO
(3) Nitric oxide into N_2
(4) CO_2 into hydrocarbons

73. Choose the incorrect match.
(1) Bone – Hard and non – pliable ground substance rich in Ca^{2+} and collagen fibres
(2) Platelets – Enucleated and the smallest formed elements.
(3) Red bone marrow – Diaphysis of long bone
(4) Cartilage – Outer ear joints, tip of nose

74. Euro III norms stipulate that sulphur be controlled at ___ ppm in diesel, ___ ppm in petrol and aromatic hydrocarbons are to be contained at ___ percent of the concerned fuel.
(1) 350, 150, 42 (2) 150, 350, 35
(3) 350, 150, 50 (4) 150, 350, 42

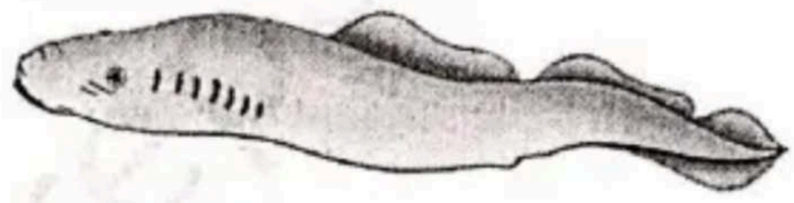
75. Biochemical Oxygen Demand (BOD) in a river water
(1) has no relationship with concentration of O_2 in the water
(2) remains unchanged when biodegradable pollutants get mixed with a lake water.
(3) decreases when sewage gets mixed with river water
(4) increases when sewage gets mixed with river water

76. Study the following statements
I. Recycling carried out in an environment friendly manner is the only solution for the treatment of e – wastes
II. Polyblend when used to lay roads, increased the road life, by enhancing the bitumen's water repellant properties
III. There is a danger of seepage of chemicals from sanitary landfills, polluting underground water resources
Which of the above are correct?
(1) I and II only (2) II and III only
(3) I and III only (4) I, II, III

77. Given below are two statements
Statement-I (S-I): Thin walled urinary bladder is present dorsal to the rectum in frog.
Statement-II (S-II) : A pair of renal portal veins carry deoxygenated blood from lower parts of the body to kidneys..
In the light of the above statements, choose the most appropriate answer from the options given below
(1) S-I and S-II are true
(2) S-I and S-II are false
(3) S-I is true, but S-II is false
(4) S-I is false, but S-II is true

78. Choose the incorrect match about cockroach
(1) Gonapophyses in male – Chitinous and asymmetrical
(2) Each ovary – Eight ovarioles
(3) The next to last nymphal stage - Wings
(4) Development –Paurometabolous

79. Identify the following figure.



Which of the following is not related to the figure?
(1) Indirect development
(2) 6 – 15 pairs of gill slits
(3) 2 pairs of paired fins
(4) cartilaginous cranium

80. Match the following

A) Walls of blood vessels and alveoli of lungs	i) Ciliated epithelium (columnar or cuboidal cells)
B) Ducts of gland and tubular parts of nephron.	ii) Non-Ciliated simple columnar epithelium
C) Stomach and intestine	iii) Simple cuboidal epithelium
D) Bronchioles and fallopian tubes	iv) Simple squamous epithelium

(1) A – ii, B – i, C – iii, D – iv
(2) A – iv, B – iii, C – ii, D – i
(3) A – iii, B – iv, C – ii, D – i
(4) A – ii, B – iii, C – i, D – iv

81. In earthworm, removal of spermathecae results in.
(1) Failure of reduction division during spermatogenesis
(2) Failure of storage of sperms of its own
(3) Failure of spermatogonia production
(4) Failure of storage of sperms of other earthworm

82. Find out the number of segments that contain only exonephric nephridia in *Pheretima*.
(1) 6 (2) 7
(3) 8 (4) 9

83. Heart of cockroach is differentiated into funnel shaped chambers with ___(A) ___ on either side. Blood from ___(B) ___ enter ___(C) ___ through ___(D) ___ and is pumped ___(E) ___ to sinuses again.
In the above, A, B, C, D and E are respectively.
(1) Sinuses, heart, sinuses, stigmata and posteriorly
(2) Ostia, sinuses, heart, ostia and anteriorly
(3) Ostia, heart, sinuses, ostia and posteriorly
(4) Ostia, sinuses, heart, ostia and posteriorly

84. Study the following statements about proventriculus in cockroach
a) It is the first part of the mesenteron
b) It secretes amylases, proteases and lipases
c) It is lined by endoderm
d) It has an outer layer of thick circular muscles and thick inner cuticle forming six highly chitinous plate called teeth.
Which of the above are correct?
(1) a and c (2) c and d
(3) b and d (4) d only

85. Find out the incorrect one regarding frog.
(1) Ovisacs in female open into cloaca
(2) There is no functional connection between ovaries and kidneys.
(3) Cloaca is used to pass faecal matter, urine and sperms in male
(4) Testes are found adhered to the upper part of kidneys by mesovarium.

86. Match the following w.r.t. cockroaches.

A) $2^{nd} - 6^{th}$ abdominal segments	i) Testes
B) 6th abdominal segment	ii) Mushroom gland
C) $6^{th} - 7^{th}$ abdominal segments	iii) Ovaries
D) $4^{th} - 6^{th}$ abdominal segments	iv) Spermatheca

(1) A - i, B – iv, C – iii, D – ii
(2) A – iii, B – iv, C – ii, D – i
(3) A – ii, B – iii, C – i, D – iv
(4) A – i, B – iv, C – ii, D - iii

87. In neural tissue of human.
(1) Bipolar neuron has two dendrites.
(2) Nissl granules are present in dendrites and axon of neuron.
(3) Schwann cells form myelin sheath around axon in CNS
(4) Neuroglia make up more than one half the volume.

88. Muscle fibres of biceps muscle are characterized by.
(1) Fusiform and voluntary.
(2) Highly branched and voluntary.
(3) Intercalated discs for rapid conduction of impulses.
(4) Multinucleated condition and striated.

89. Match the following

A) Dry surface of skin	i) Transitional epithelium
B) Moist surface of buccal cavity and pharynx	ii) Stratified cuboidal epithelium
C) Ducts of salivary glands and pancreas	iii) Non-Keratinised stratified squamous epithelium
D) Urinary bladder and ureters	iv) Keratinised stratified squamous epithelium

(1) A – iii, B – ii, C – i, D – iv
(2) A – iv, B – ii, C – iii, D – i
(3) A – iv, B – iii, C – ii, D – i
(4) A – iv, B – iii, C – i, D – ii

90. Match the following:

A) Tendons and ligaments	i) Areolar connective tissue
B) Dermis of skin	ii) Dense irregular tissue
C) Subcutaneous layer	iii) Areolar and adipose connective tissue
D) Supporting framework for epithelium	iv) Dense regular connective tissue

(1) A – ii, B – iv, C – i, D – iii
(2) A – iv, B – ii, C – i, D – iii
(3) A – i, B – ii, C – iv, D – iii
(4) A – iv, B – ii, C – iii, D – i

91. Chipko movement of Garhwal Himalayas is related to
(1) Protecting the environment from the water pollutants
(2) Protecting the soil from e-wastes
(3) Protecting trees from the axe of contractors
(4) Protecting ozone layer from the ozone depleting substances

92. Study the following statements w.r.t. cockroach
a) It can receive several images of an object with the help of several ommatidia.
b) Kind of vision is mosaic vision with less sensitivity but more resolution.
c) If the head of cockroach is cut off, it will still live for as long as one week.
d) It has ten pairs of segmental ganglia
Which of the above are incorrect?
(1) b and d (2) c and d
(3) a and d (4) a and b

93. Choose the incorrect match
(1) Gap junctions – Connecting cytoplasm of adjoining cells.
(2) Connective tissue – Most abundant and widely distributed in the body.
(3) Compound epithelium – Protection against chemical and mechanical stresses.
(4) Adhering junctions – Stop substances from leaking across a tissue.

94. Which of the following is the correct representation of pancreatic enzymes
(1) Dipeptidase, amylase, lipase
(2) Trypsinogen, carboxypeptidase, nucleosidase
(3) Amylase, Nuclease, chymotrypsinogen
(4) Lipase, Nucleotidase, Maltase

95. The following characters are related to phylum
a) Aquatic, mostly marine
b) Sessile or free swimming
c) Radially symmetrical animals
d) Some of them show polymorphism
(1) Porifera (2) Nematoda
(3) Echinodermata (4) Coelenterata

96. Considering the gross calorific values, which food item is most effectively utilized in the body
(1) Carbohydrates (2) Fats
(3) Proteins (4) Roughage

97. Given below are two statements
Statement-I (S-I): Tapeworm, roundworm and pinworm are endoparasites of human intestine.
Statement-II (S-II) : All endoparasites of human are unisexual.
In the light of the above statements, choose the most appropriate answer from the options given below
(1) S-I and S-II are true
(2) S-I and S-II are false
(3) S-I is true, but S-II is false
(4) S-I is false, but S-II is true

98. Study the following statements regarding *Periplaneta*.
a) A pair of vas deferens opens into ejaculatory duct through seminal vesicles.
b) The sperms are stored in the seminal vesicles and are glued together in the form of bundles called spermatophores.
c) Layers of spermatophore capsule are formed by the secretions of utriculi majors, ejaculatory duct and phallic gland.
d) A pair of spermathecae in female open into vagina.
Which of the above are correct?
(1) c and d (2) a, c and d
(3) a, b and c (4) b, c and d

99. Given below are two statements
Statement I: Largest gland of the body is situated just below the diaphragm, towards left side in the abdominal cavity
Statement II: Second largest gland is a compound gland and is situated between the limbs of the C shaped duodenum
In the light of the above statements, choose the most appropriate answer from the options given below
(1) S-I and S-II are true
(2) S-I and S-II are false
(3) S-I is true, but S-II is false
(4) S-I is false, but S-II is true

100. Given below are two statements: one is labelled as Assertion (A) and other is labelled as Reason (R).
Assertion (A): Nuclear waste is considered as an extremely potent pollutant
Reason (R): At high doses, nuclear radiation is lethal but at lower dose, it creates various disorders, the most frequent of all being cancer
In the light of the above statements, choose the correct answer from the options given below
(1) A is correct but R is not correct
(2) A is not correct but R is correct
(3) Both A and R are correct and R is the correct explanation of A
(4) Both A and R are correct but R is not the correct explanation of A

PHYSICS

SECTION – A

101. A man starts walking at t=0 on a straight road from his home to a market 2.5 km away with a speed of 5 kmh^{-1}. Finding the market closed he instantly walks back home with 7.5 kmh^{-1}. The average speed of the man over the interval 0 to 50 min is
(1) 6 kmh^{-1} (2) 1.25 kmh^{-1}
(3) zero (4) 2.5 kmh^{-1}

102. A car moving along a straight highway with a speed of 90 kmh^{-1} is brought to rest by applying brakes. If brakes cause a uniform retardation and stops within a distance of 125m, how long did it take to stop the car ?
(1) 5 s (2) 10 s
(3) 1.38 s (4) 8 s

103. A body thrown vertically up from ground is at a point 'P' after 4s and 6s of projection. Time of ascent of of the body is
(1) 10s (2) 5s
(3) 2s (4) 3s

104. Position of a particle is given by $\vec{r} = \left(3.0t\,\hat{i} - 2.0\,t^2\hat{j} + 40\hat{k}\right)$m. Velocity of the particle at t = 1s is
(1) 4 ms^{-1} along x-axis
(2) 5 ms^{-1}, at 53^0 with x-axis
(3) 5 ms^{-1} at 37^0 with x-axis
(4) 3 ms^{-1} at 53^0 with x-axis

105. It is raining vertically with a speed of 5 ms^{-1}. A woman rides a bicycle with a speed of $5\sqrt{3}\ ms^{-1}$ in north to south direction. The angle at which she has to hold umbrella is
(1) 30^0 with horizontal towards north
(2) 60^0 with horizontal towards north
(3) 30^0 with horizontal towards south
(4) along vertical

106. A cricketer can throw a ball to a maximum horizontal distance of 100m. The maximum height to which he can throw the ball is
(1) 50 m (2) 100 m
(3) 200 m (4) 150 m

107. Assertion (A) : For a particle in uniform circular motion, the acceleration averaged over one cycle is a null vector.
Reason (R) : For a particle in uniform circular motion velocity is variable
(1) (A) and (R) are true and (R) is the correct explanation of (A)
(2) (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true (R) is false
(4) (A) is false (R) is true

108. Which of the following is not correct ?
A) If bullet fired at 30^0 with horizontal can hit a target 3 km away, by adjusting the angle it can be made to hit a target 5 km away.
B) $\left|\vec{a}+\vec{b}\right| = \left|\vec{a}\right| + \left|\vec{b}\right| \Rightarrow \vec{a}$ and $\vec{b}$ are collinear
(1) neither A nor B (2) Only A
(3) Both A and B (4) only B

109. If the magnitude of the resultant of two vectors of the same magnitude is twice the magnitude of their difference, then what is the angle between these vectors.

(1) 37^0 (2) 45^0

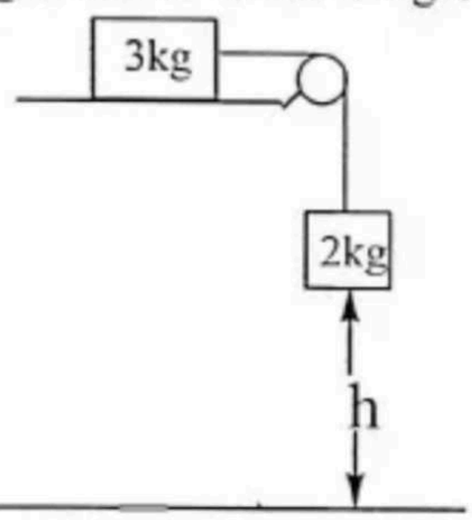
(3) 53^0 (4) 60^0

110. A marble projected horizontally with a velocity 'u' from the top most step of a stair case. Width and height of each step is are 30 cm and 20 cm respectively. If it hits the edge of 4th step then 'u' is ($g=10\ ms^{-2}$)

(1) 4 (2) 2

(3) 5 (4) 3

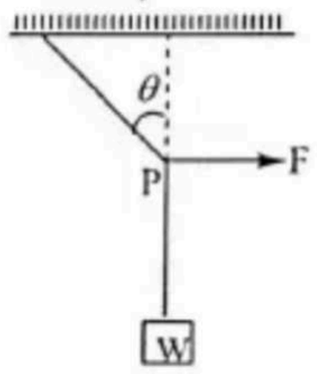
111. 2 kg and 3 kg blocks are connected by a string as shown. Initially 2 kg block is at a height of 2m above the ground . On releasing the system the time taken by 2kg block to strike the ground is



(1) 1 s (2) 2 s

(3) 0.5 s (4) 0.25s

112. A block of mass 3kg is suspended by a rope of length 2m from the ceiling. A force 40 N in the horizontal direction is applied at the mid point P of the rope as shown. The angle made by the rope with vertical in equilibrium is ($g = 10\ ms^{-2}$ and $\sin 37^0 = 3/5$)



(1) 37^0 (2) 53^0

(3) $\sin^{-1}\left(\frac{3}{4}\right)$ (4) $\cos^{-1}\left(\frac{3}{4}\right)$

113. A bullet of mass 40 g moving with a speed of $90ms^{-1}$ enters a heavy wooden block and is stopped after a distance of 60 cm what is the average resistive force exerted by the block on bullet ?

(1) 540 N (2) 270 N

(3) 135 N (4) 240 N

114. A hiker stands on the edge of a cliff 495m above ground and throws a stone horizontally with a speed 'u'. If it strikes the ground with a speed 100 ms^{-1} then 'u' is

(1) 10 ms^{-1} (2) 4.95 ms^{-1}

(3) 50 ms^{-1} (4) 20 ms^{-1}

115. Assertion (A) : If we look out of a window of a fast moving train, the nearby trees seem to move rapidly, but the distant hill tops seem to be stationary.

Reason (R) : The angle made by far off objects for the observer is small and if he moves the change in angle is also very small.

(1) A and R are true, R is correct explanation of A

(2) A and R are true, R is not correct explanation of A

(3) A is true, R is false

(4) A is false, R is true

116. The distance of the sun from earth is 1.5×10^{11} m and its angular diameter is (2000) s when observed from the earth. The diameter of the sun will be

(1) 2.45×10^{10}m (2) 1.45×10^{10} m

(3) 1.45×10^{9} m (4) 0.14×10^{9} m

117. Identify the term having dimensions of time L = inductance, C = capacitance, R = resistance

(1) L/R (2) CR

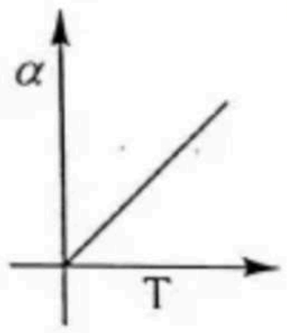
(3) $\sqrt{LC}$ (4) all of these

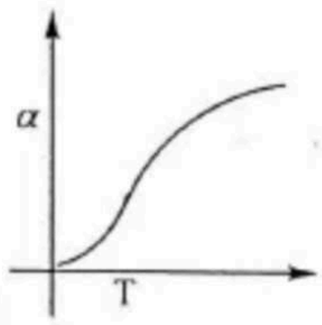
118. An iron rim of diameter 2.255m at 25^0C is to be fixed on a wooden wheel of diameter 2.250m. Coefficient of linear expansion of iron is $12\times10^{-6}\ {}^0C^{-1}$. Temperature to which the rim is to be heated is

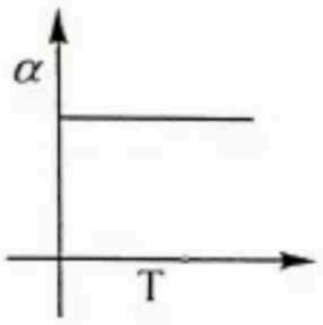
(1) 185^0C (2) 210^0C

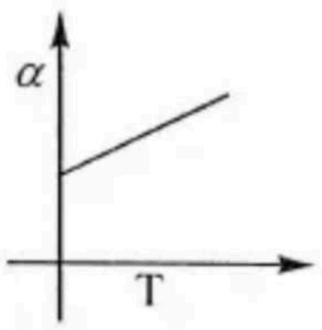
(3) 160^0C (4) 200^0C

119. Variation of coefficient of volume expansion of copper with temperature

(1) 

(2) 

(3) 

(4) 

120. Triple point of carbon dioxide is 216.55 K. Its value in Fahrenheit is

(1) -69.88^0F (2) -56.60^0F

(3) -169.88^0F (4) -156.60^0F

121. When a piece of iron is heated in a hot flame, it first becomes dull red, then reddish yellow and finally white hot. This phenomenon can be explained by

(1) Stefan – Boltaman's law

(2) Green house effect

(3) Wien's displacement law

(4) Newton's law of cooling

122. Three rods A,B and C of same length and same area of cross section are connected in series. Their thermal conductivity are K, 2K and 3K respectively. If the temperature difference across the free ends of the composite rod is 55^oC then temperature difference across the rod B is

(1) 30^oC (2) 10^oC

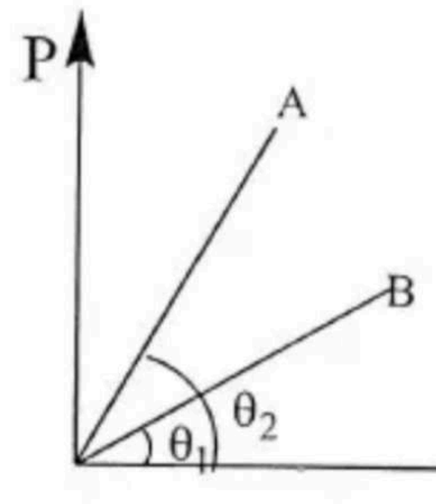
(3) 15^oC (4) 11^oC

123. Radiant power received by a plate arranged at a distance 'd' from a point source is 'P'. If temperature of the source and distance of the plate from source are halved then the radiant power received by the plate would be

(1) P (2) $\frac{P}{4}$

(3) 4P (4) 128P

124. Graphs drawn between pressure (P) and absolute temperature (T) of two samples of gases of same volume are as shown. If sample A is 1g hydrogen then the sample B is $\left(\theta_2 = 2\theta_1 = 60^0\right)$



(1) 1.5g Helium (2) 2g Helium

(3) (2/3)g Helium (4) 3g Helium

125. A closed vessel contains 2g Helium gas at a temperature 27^0. To double the r.m.s speed of its molecules the temperature of the gas is to be raised by

(1) 900 K (2) 81^0C

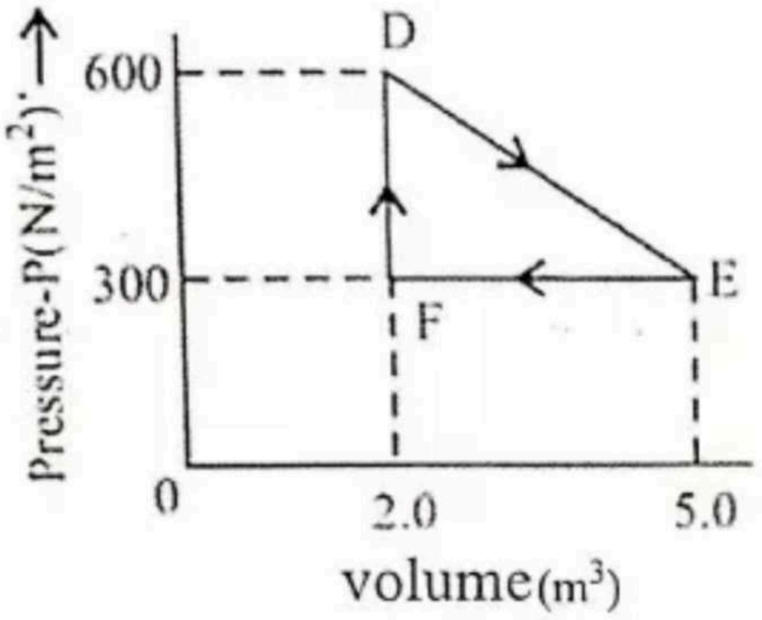
(3) 1200 K (4) 1200^0C

126. Water of volume 3 L in a closed container is heated with a coil of 1 kW. While water is heated, the container loses energy at a rate of 125 J/s. In how much time will the temperature of water rise from 27^0C to 77^0C ? (Specific heat of water is 4.2 kJ/kg and that of the container is negligible).

(1) 8 min 20 s (2) 6 min 2 s

(3) 7 min 2s (4) 12 min

127. A thermodynamic system is taken from an original state D to an intermediate state E by the linear process shown in the figure. Its volume is then reduced to the original volume from E to F by an isobaric process. The total work done by the gas from D to E to F will be



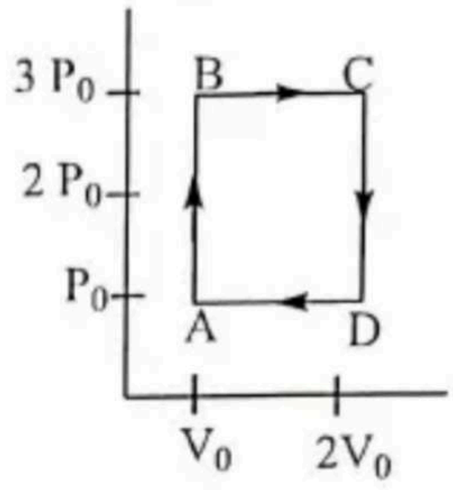
(1) −450 J (2) 450 J
(3) 900 J (4) 1350 J

128. A sample of a certain diatomic ideal gas undergoes a temperature increase of 40 K at constant pressure. If heat given to the sample is 70 cal work done by the gas in this process is (given $R = 8.3\ JK^{-1}\ mol^{-1}$)
(1) 20 cal (2) 20 J
(3) 50 cal (4) 50 J

129. An electric heater supplies heat to a system at a rate of 23.8 cal s^{-1}. If the system per forms work at a rate of 75 J s^{-1} then the rate at which its internal energy increases is nearly
(1) 345 W (2) 25 W
(3) 125 W (4) 495 W

130. A reversible engine has an efficiency of 1/4. If the temperature of the sink is reduced by 100^0C, its efficiency becomes double. Calculate the initial temperature of the sink.
(1) 300 K (2) 300^0C
(3) 400 K (4) 400^0C

131. An engine operates by taking a monoatomic ideal gas through the cycle shown in the figure. Total heat change involved in the process is



(1) $+P_oV_o$ (2) $--P_oV_o$
(3) $+2P_oV_o$ (4) $--2P_oV_o$

132. Match the following.

Set – I	Set-II
A. Regelation	p. Measuring rods / scales
B. Anamolous behavior of water	q. Pressure cooker
C. Low value of coefficient of expansion of invar steel	r. Snow skating
D. Increase in boiling point with increase in pressure	s. Existance of aquatic life in winter in cold countries

(1) A-r; B-s; C-q; D-p
(2) A-r; B-s; C-p; D-q
(3) A-s; B-r; C-p; D-q
(4) A-p; B-s; C-r; D-q

133. Assertion (A) : In modern thermometry, triple point of water is taken as standard fixed point
Reason (R) : Melting point of ice and boiling point of water depend on pressure.
(1) (A) and (R) are true and (R) is the correct explanation of (A)
(2) (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true (R) is false
(4) (A) is false (R) is true

134. A copper block of mass 2.5 kg is heated in a furnace for a long time and dropped in cavity of a large ice block. Specific heat of copper is 0.4 J g^{-1} C^{-1}. If 1.5 kg of ice melts on cooling of copper block to 0^0C, then temperature of furnace is

(1) 504 K (2) 504^0C

(3) 405^0C (4) 405K

135. A black body is at a temperature of 2800 K. The energy of radiation emitted by this object with wavelength between 499 nm and 500 nm is U_1, between 999 nm and 1000 nm is U_2 and between 1499 nm and 1500 nm is U_3. The Wien's constant 2.88 x 10^6 nm K. Then

(1) $U_1 > U_2$ (2) $U_3 > U_2$

(3) $U_1 > U_2 < U_3$ (4) $U_1 < U_2 > U_3$

SECTION-B

136. Two masses 'M' and 12 kg are connected at the ends of a light inextensible string that goes over a fixed smooth pulley. If blocks are initially at same level and the separation between the blocks after 1s of releasing is 2m then 'M' is (g=10 ms^{-2} and M < 12kg)

(1) 4 kg (2) 8 kg

(3) 3 kg (4) 6 kg

137. A car of mass m starts from rest and acquires a velocity along east $\upsilon = \upsilon\hat{i}\,(\upsilon > 0)$ in two seconds. Assuming the car moves with uniform acceleration, the force exerted on the car is

(1) $\frac{m\upsilon}{2}$ eastward and is exerted by the car engine

(2) $\frac{m\upsilon}{2}$ eastward and is due to the friction on the tyres exerted by the road

(3) more than $\frac{m\upsilon}{2}$ eastward exerted due to the engine and overcomes the friction of the road

(4) $\frac{m\upsilon}{2}$ exerted by the engine

138. A player throws a ball upwards with an initial speed of 25 ms^{-1}. Average velocity of ball in first 3s is (g=10 ms^{-2})

(1) 10 ms^{-1} (2) 20 ms^{-1}

(3) 12 ms^{-1} (4) 8 ms^{-1}

139. A particle of mass 100 g starts from origin at t = 0 with a velocity $5\hat{i}\ ms^{-1}$ and moves in xy plane under the action of a force $\left(0.3\,\hat{i} + 0.2\hat{j}\right)$N. When its y-coordinate is 36 m, its x-coordinate is

(1) 36 m (2) 48 m

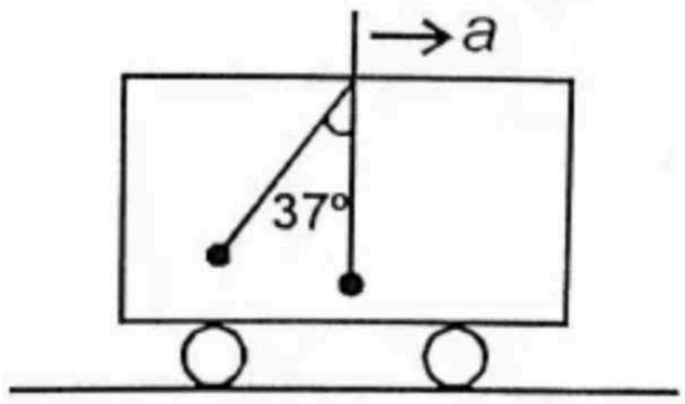
(3) 84 m (4) 30 m

140. For an oblique projectile, the ratio of de Broglie wavelength at its highest point to that at point of projection is 2. The ratio of its horizontal range to maximum height is

(1) $4\sqrt{3}$ (2) $\frac{1}{4\sqrt{3}}$

(3) $\frac{4}{\sqrt{3}}$ (4) $\frac{\sqrt{3}}{4}$

141. A pendulum with a bob of mass 2 kg is hanging from the ceiling of a trolley as shown in the figure. The trolley is uniformly accelerating so that the bob moves backwards as shown then the pseudo force on the bob is ($g = 10$ m/s^2)



(1) zero (2) 7.5N

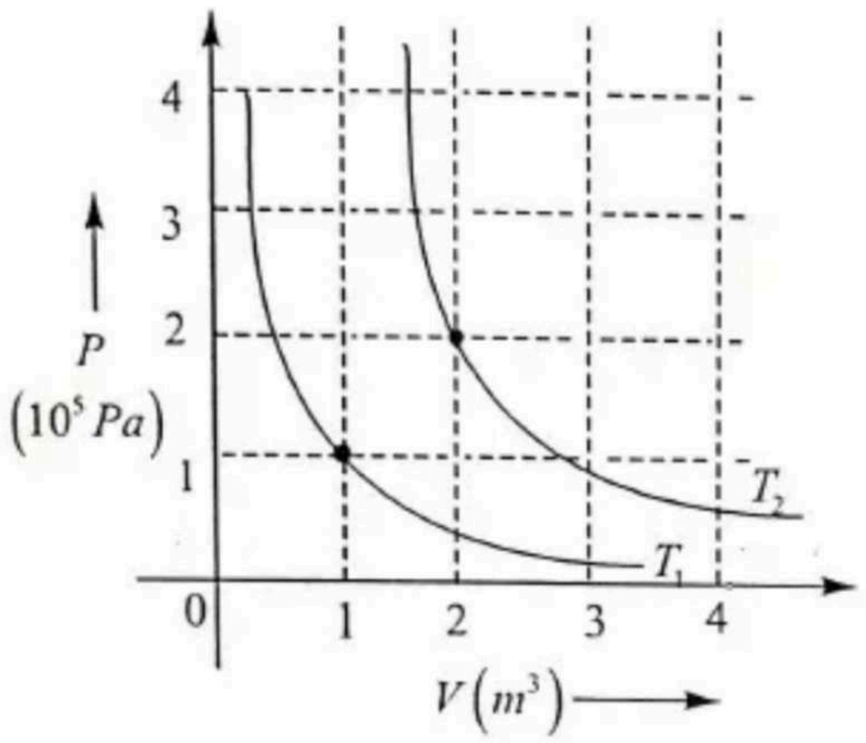
(3) 15N (4) 2.5N

142. A block of mass m is at rest on a rough inclined plane of inclination θ. The force exerted on plane by the block is

(1) $mg\sin\theta$ (2) $mg\cos\theta$

(3) mg (4) Zero

143. The following graph shows two isotherms for a fixed mass of an ideal gas. The ratio of r.m.s speed of the molecules at temperatures T_1 and T_2 is



(1) $\sqrt{2}$ (2) $\frac{1}{2}$

(3) 2 (4) 4

144. A cylinder with a movable piston contains 3 moles of hydrogen at standard temperature and pressure. The walls of the cylinder are made of a heat insulator, and the piston is insulated by having a pile of sand on it. If the gas is compressed to half its original volume then

a) pressure of the gas increases

b) temperature the gas remains same

c) internal the gas increases

(1) a & c are correct

(2) a & b are correct

(3) b & c are correct

(4) all the given are correct

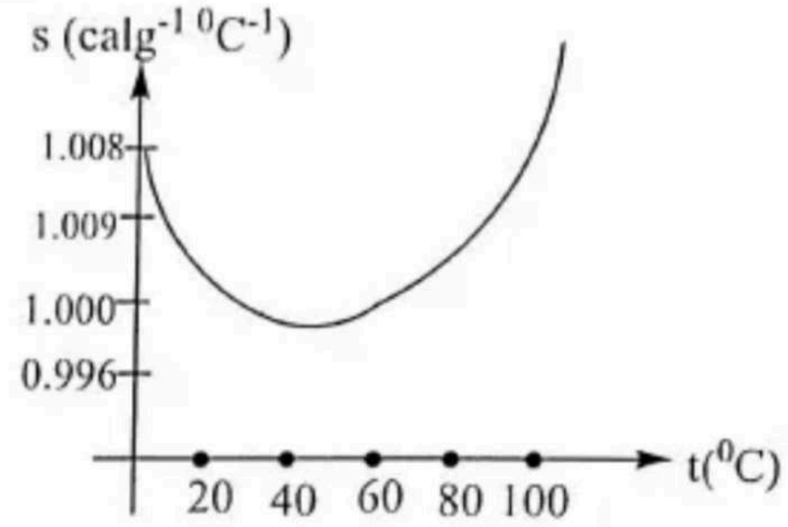
145. A diatomic gas is used in a Carnot engine as the working substance. If during the adiabatic expansion part of the cycle the volume of the gas increases from V to 32V, the efficiency of the engine is

(1) 0.25 (2) 0.5

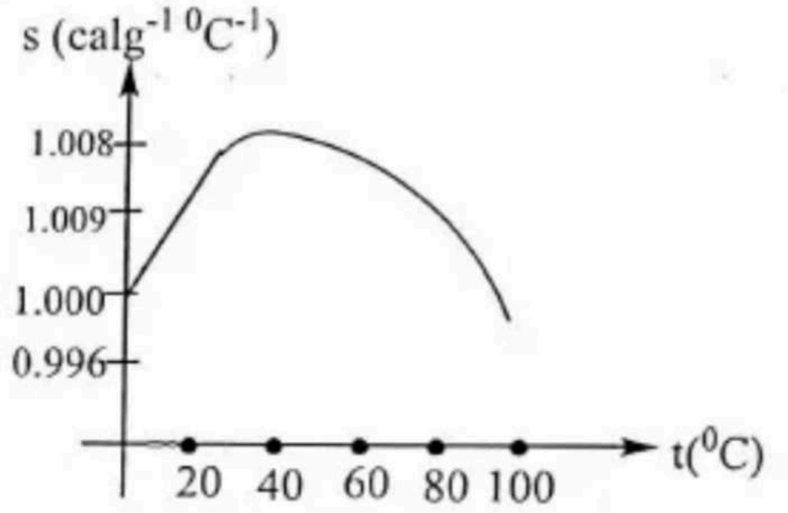
(3) 0.75 (4) 0.99

146. Variation of specific heat of water (s) with temperature (t) is

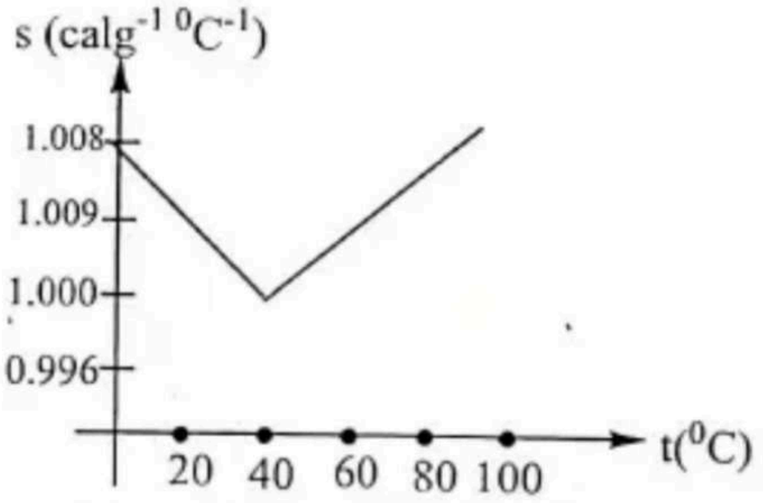
(1)



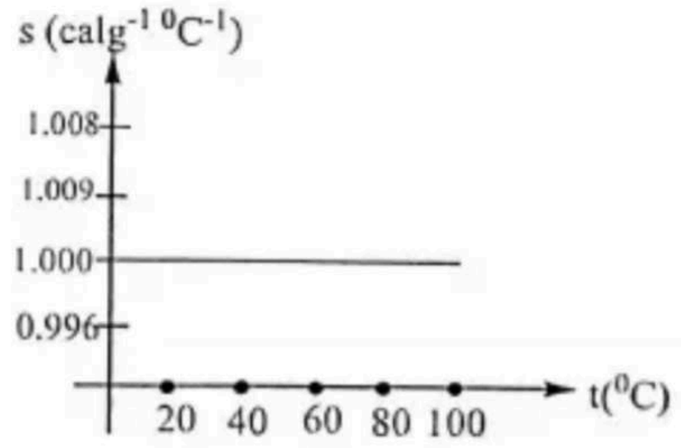
(2)



(3)



(4)

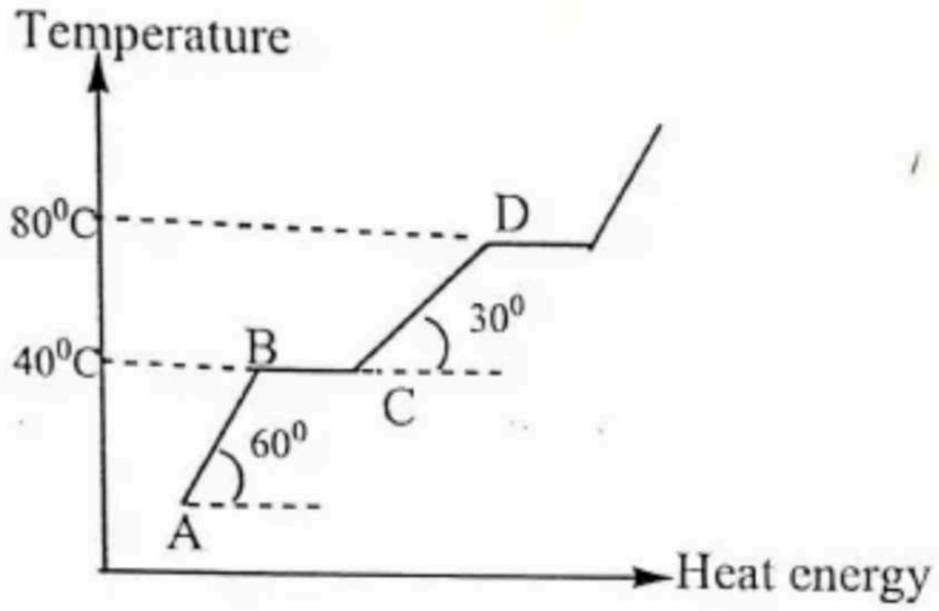


147. Two rods A and B of identical dimensions are at temperature 30^0C. If A is heated upto 180^0C and B upto T^0C, then the new lengths are the same. If the ratio of the coefficients of linear expansion of A and B is 4:3, then the value of T is

(1) 230^0C (2) 270^0C

(3) 200^0C (4) 250^0C

148. A plot of temperature versus heat energy of substance initially in solid state is as shown. If the specific heat of solid is 0.2 cal g^{-1} $^0C^{-1}$, then the amound if heat required to rise 50 g of liquid from 40^0C to its boiling point C is



(1) 600 cal (2) 600 J

(3) 200 J (4) 1200 cal

149. A pendulum clock gives correct time at 20^0C at a place where g=10 m/s^2. The pendulum consists of light steel rod connected to a heavy ball. If it is taken to different place where g=10.01 m/s^2 at what temperature the pendulum gives correct time ($\alpha_{steel}^{\cdot} = 10^{-5} / {}^0C$)

(1) 30^0C (2) 100^0C

(3) 60^0C (4) 120^0C

150. Assertion (A) : The climate of harbor town is more temperate than a town in a desert at same altitude.

Reason (R) : Relative humidity in harbor town is greater than that in desert town.

(1) (A) and (R) are true and (R) is the correct explanation of (A)

(2) (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true (R) is false

(4) (A) is false (R) is true

CHEMISTRY
SECTION – A

151. (A) Total number of electrons with n = 4, m = -1 are 'X'.
(B) Number of electrons with (n + *l*) = 3 in magnesium atom are 'Y'.
Difference of magnitude of between 'X' and 'Y' is

(1) 0 (2) 2
(3) 4 (4) 5

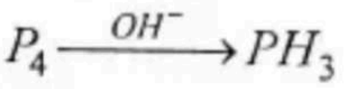
152. IUPAC name of



$CH_2 - C \equiv C - CH = CH - CH_2 - CH_3$ is
|
Cl

(1) 1–chlorohept–4–ene–2–yne
(2) 6–chlorohept–3–ene–5–yne
(3) 1–chlorohept–2–ene–4–yne
(4) 6–chlorohept–5–ene–3–yne

153. Number of hydroxyl ions involved in the following balanced half reaction is



$P_4 \xrightarrow{OH^-} PH_3$

(1) 4 (2) 8
(3) 12 (4) 6

154. At high pressure for one mole of a gas a graph is plotted between 'Z; (y - axis) vs $\frac{P}{T}$ (x – axis). The slope of the graph is 0.5 k/atm. Value van der Waals constant 'b' is

(1) 0.164 lit / mole (2) 0.0205 lit/ mole
(3) 0.082 lit / mol (4) 0.041 lit / mole

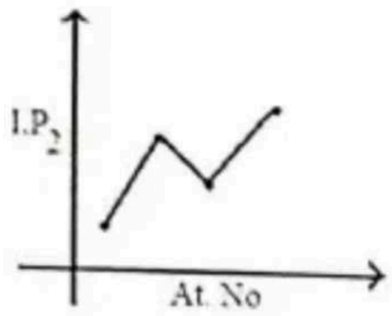
155. Correct match of the pair of compounds given in column II with column I is

Column – I	Column – II
A) Metamers	i) Catechol and resorcinol
B)Positional isomers	ii) Cresol and benzyl alcohol
C)Functional isomers	iii) Pentan–2-one and pentan-3-one
D) Homologues	iv) Pent-2-yne and 3-methyl – but–1- yne
E) Chain isomers	

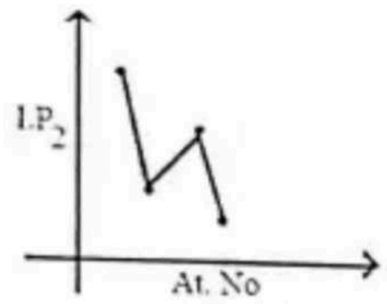
(1) A – iii, B – i, C – ii, E – iv
(2) B – iii, B – i, C – ii, E – iv
(3) A – iii, B – iv, C – i, D – ii
(4) A – iii, B – iv, C – ii, D - i

156. Correct graphical representation of I.P2 of Li, Be, B and C is

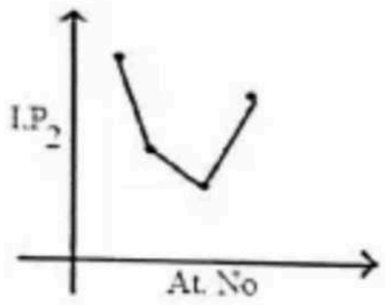
(1)



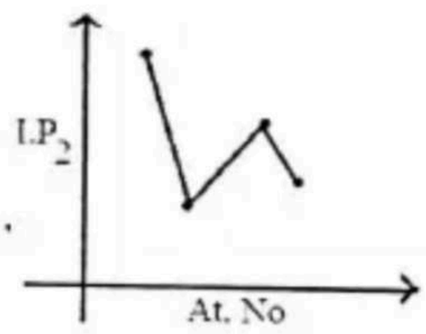
(2)



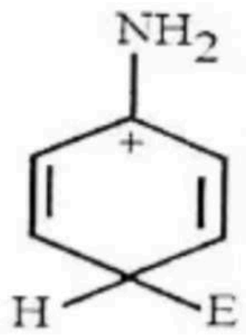
(3)

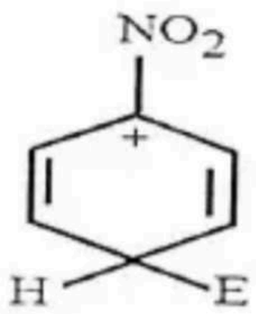


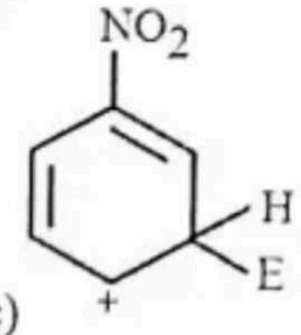
(4)

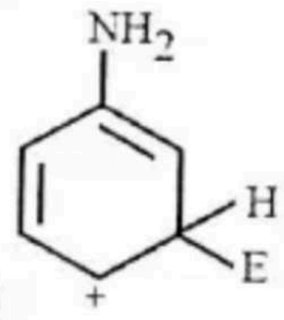


157.

(a) 

(b) 

(c) 

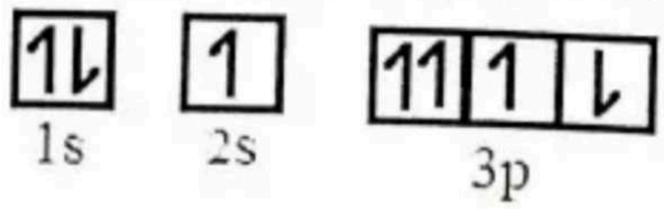
(d) 

Most stable and least stable arenium ions are respectively

(1) a, c (2) a, b

(3) b, d (4) c, d

158. The following configuration violates



(1) Aufban's principle, Pauli's principle and Hund's rule

(2) Aufban's principle and Pauli's principle only

(3) Pauli's principle only

(4) Aufban's principle and Hund's rule only

159. Number of carbon atoms present in 12 amu of carbon is

(1) 6.023×10^{23} (2) 6.023×10^{24}

(3) 1 (4) 6

160. Best – method to purify impure aniline is

(1) Vacuum distivation

(2) Fractional distillation

(3) Simple distillation

(4) Steam distillation

161. Correct match is

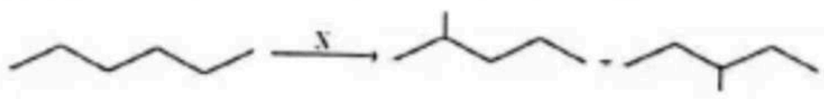
Column – I (Element)	Column – II (Symbol)
A) Hassnium	i) Uup
B) Meitnerium	ii) Unb
C) Moscovium	iii) Uno
D) Copernicium	iv) Une

(1) A – iv, B – iii, C – ii, D – i

(2) A – iii, B – iv, C – i, D – ii

(3) A – ii, B – iii, C – iv, D – i

(4) A – ii, B – iv, C – i, D – iii

162. 

The reagent /catalyst in this reaction is

(1) V_2O_5/773k, 10-20 atm

(2) Mo_2O_3 / Δ

(3) Cu/573k, 100 atm

(4) HCl / anhydrous $AlCl_3$

163. 14 grams of an ideal gas occupies 5.6 liters at 2 atmospheres and $273K$. The gas would be

(1) N_2 (2) SO_2

(3) C_4H_8 (4) C_4H_{10}

164. Correct decreasing order of ionic size among the following is

(1) $Cs^+ > Cl^- > H^-$ (2) $Cl^- > H^- > Cs^+$

(3) $H^- > Cl^- > Cs^+$ (4) $Cl^- > Cs^+ > H^-$

165. A compound contain 10.04 %, 0.837 % and 89.12% by weight of carbon, hydrogen and chlorine respectively. Empirical formula of the compound is

(1) $CHCl_3$ (2) CH_2Cl_3

(3) CH_3Cl (4) C_2HCl_5

166. 1-butene $\xrightarrow[CCl_4]{Br_2} X$. Regarding this process, incorrect statement is

(1)A cyclic bromonium intermediate is formed

(2) A vicinal dibromide is the product obtained

(3) Rearranged product Cannot be obtained

(4) 'X' is 1,1-dibromobutane

167. The incorrect statement among the following is :

(1) There are three significant figures in 285 cms

(2) There are two significant figures in 0.25ml

(3) There are three significant figures in 0.03

(4) There are four significant figures in 2.005

168. **Assertion (A):** Ammonia can be liquefied easily than nitrogen.

Reason (R): Critical temperature of ammonia is higher than that of nitrogen.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true and ® is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

169. Which one of the following compound gives blood red colour by the addition of $FeCl_3$

(1) Thio ether

(2) Ammonium sulphide

(3) Thio urea

(4) Thio aceton

170. The orbitals having more number of angular nodes than radial nodes among the following are

(A) 3s (B) 4d

(C) 5p (D) 6f

(E) 2p

(1) B, D and E only

(2) B, C, D and E

(3) B, C and D only

(4) A, B, D and D only

171. **Statement I:** Both 2-methylbut-2-ene and 4-methylpent-2-ene give ketone as one of the product when with $KMnO_4 / H^+$.

Statement II: Alkenes undergo oxidative cleavage when treated with $KMnO_4 / H^+$

(1) Both statement I and statement II are true

(2) Statement I is true and statement II is false

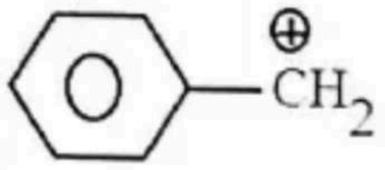
(3) Statement I is false and statement II is true

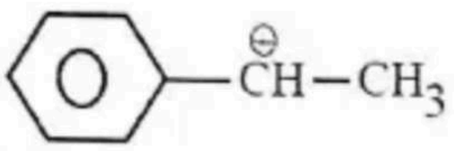
(4) Both statement I and statement II are false

172. The species which is stabilized by resonance but not by hyper conjugation is

(1) $CH_3-\overset{\oplus}{C}H_2$

(2) $CH_2=CH-\overset{\oplus}{C}H-CH_3$

(3) 

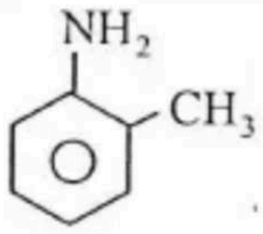
(4) 

173. 200 ml of a moist gas at 27^0C exerts a pressure of 90 mm Hg. If this gas is transferred into a 500 ml vessel at same temperature, the pressure of moist gas would be (aqueous tension at 300 k = 30 mm Hg).

(1)54 mm (2) 24 mm

(3) 69 mm (4) 45 mm

174. **Statement I:** IUPAC name of



is 2 – amino toluene

Statement II: 1 – pentyne and 2 – pentyne are chain isomers

(1) Both statement I and statement II are true

(2) Statement I is true and statement II is false

(3) Statement I is false and statement II is true

(4) Both statement I and statement II are false

175. Correct increase order of boiling point of the first three alkanes is represented in

(1) CH_4, C_2H_6, C_3H_8

(2) C_3H_8, C_2H_6, CH_4

(3) CH_4, C_3H_8, C_2H_6

(4) C_3H_8, CH_4, C_2H_6

176. The organic compound which shows both geometrical and optical isomerism is

(1) $CH_3-CHCl-CH=CH_2$

(2) $CH_2=CH-CH(OH)-CH=CHCl$

(3) $(CH_3)_2C=CH-CHBr-CH_2CH_3$

(4) $BrCH=C=C=CHBr$

177. $CH_4 \xrightarrow[\Delta]{MoO_2} A \xrightarrow[conc.HCl]{Zn-Hg} B \xrightarrow[100atm]{O_2,Cu,523K} C \xrightarrow{CH_3MgBr} D \xrightarrow[1mole]{Cl_2/h\upsilon} E \xrightarrow[dry\ ether]{Na} F$. The compound 'D' is

(1) CH_4 (2) CH_3OCH_3

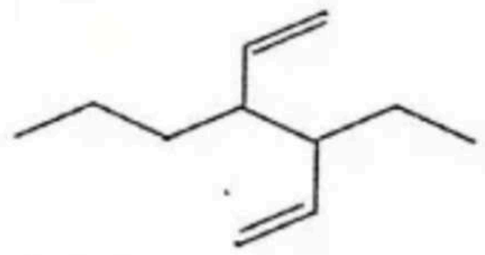
(3) C_2H_5OH (4) CH_3CHO

178. Which of the following pairs of d-orbital will have electron density along the axes?

(1) d_{z^2}, d_{xz} (2) d_{xz}, d_{y^2}

(3) $d_{z^2}, d_{x^2-y^2}$ (4) $d_{xy}, d_{x^2-y^2}$

179. The correct IUPAC name for the compound?



(1) 3-ethyl-4-propylhex-1, 5 – diene

(2) 3-propyl-4-ethylhex-1, 5-diene

(3) 3, 4-diethylhex-1,5-diene

(4) 3-Ethyl-4-propylhex-1,5-diene

180. The most suitable method for separation of 1:2 mixture of ortho and para – nitrophenols is

(1) Chromatography

(2) Crystallisation

(3) Steam distillation

(4) Sublimation

181. In an estimation of S by Carius method, 0.2175 g of the compound gave 0.5825g of $BaSO_4$ Percentage of S is

(1) 36.78 % (2) 35.50 %

(3) 36.98 % (4) 35.69 %

182. $C_6H_6 \xrightarrow[hv]{1\,mole\,Cl_2} A \xrightarrow{alc.KOH} N \xrightarrow[200°C]{H_2/Ni} C.$
Here 'C' is

(1) C_2H_6 (2) C_2H_5Cl

(3) C_2H_4 (4) C_4H_{10}

183. $CH_3-CH_2-C\equiv CH \xrightarrow{HCl} A \xrightarrow{HI} B$
Predict product B in the reaction

(1) $CH_3-CH_2-CH_2-\underset{Cl}{\overset{I}{C}}-H$

(2) $CH_3-CH_2-\overset{I}{C}H-CH_2-Cl$

(3) $CH_3-CH_2-\underset{Cl}{\overset{I}{C}}H-CH_3$

(4) $CH_3-\underset{Cl}{C}H-CH_2-CH_2-I$

184. Which of the following substituents has +M (mesomeric) effect ?

(1) $-CN$ (2) $-CHO$

(3) $-\ddot{N}H_2$ (4) $-NO_2$

185. Reductive ozonolysis of products of 2 – methyl – 2 – butene are

(1) 2 moles of HCHO

(2) $CH_3-CHO+HCHO$

(3) $CH_3-CHO+CH_3-CO-CH_3$

(4) $H-CHO+CH_3-CO-CH_3$

SECTION – B

186. RMS velocity of Helium at 400k and 2 atm is ___ m/s

(1) 1.58×10^4 (2) 158×10^3

(3) 1.58×10^3 (4) 1.58×10^2

187. In hydrogen atom, the kinetic energy of electron is 3.4 eV. The distance of that electron from the nucleus is

(1) $2.116\,A^\circ$ (2) $0.529\,A^\circ$

(3) $1.587\,A^\circ$ (4) $21.16\,A^\circ$

188. Wavelength of series limit in Balmer series of He^+ ion is

(1)91.2 nm (2) 912 nm

(3) $9.12A^o$ (4) $9120A^o$

189. **Assertion (A):** Acetic acid cannot exhibit functional isomerism.

Reason (R): Ester with two carbons is not possible.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true and ® is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

190. Exothermic process among the following is

(1) $O(g)\rightarrow O^{2-}(g)$

(2) $S^-(g)\rightarrow S^{2-}(g)$

(3) $Mg^{2+}(g)\rightarrow Mg^+(g)$

(4) $Ne(g)\rightarrow Ne^-(g)$

191. Minimum number of atoms are present in one gram of

(1) Copper (2) Chlorine

(3) Hydrogen (4) Helium

192. **Assertion (A):** Thermal decomposition of potassium chlorate cannot be called as a disproportionation reaction.

Reason(R): Potassium chlorate undergoes both oxidation and reduction.

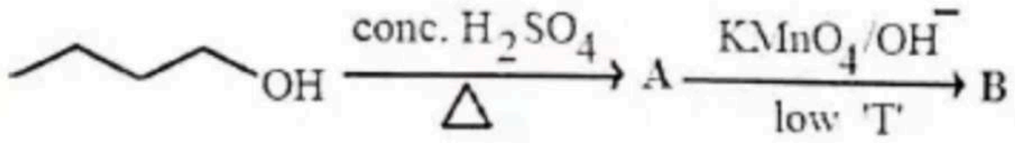
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true and ® is not the correct explanation of (A)

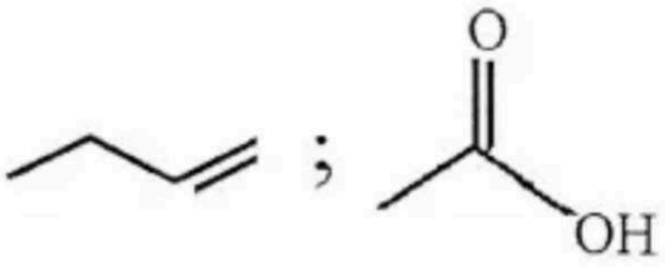
(3) (A) is true but (R) is false

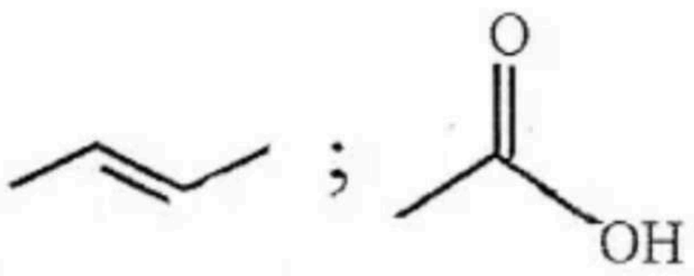
(4) Both (A) and (R) are false

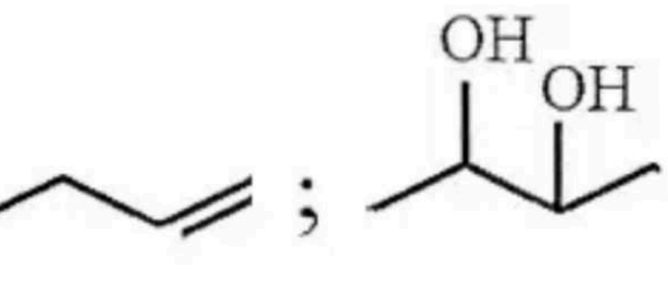
193.

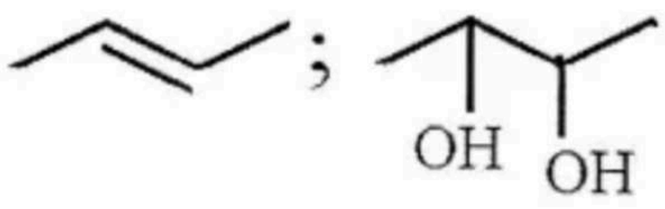


A and B are respectively

(1) 

(2) 

(3) 

(4) 

194. Identify incorrect statement from the following.

(1) Black body radiation produce continuous spectrum

(2) The four lines visible in Blamer series are red, green, blue and violet

(3) High intensity red light when incident on potassium atom can produce photo electric effect

(4) Electromagnetic radiation can pass through vacuum

195. Negative inductive effect is maximum for

(1) $-CN$ (2) $-SO_3H$

(3) $-NO_2$ (4) $-\overset{+}{N}H_3$

196. 2 grams of an organic compound in Kjeldahl's method liberated NH_3 which required 20 ml of 0.75 M H_2SO_4 solution for complete neutralization. Percentage of nitrogen in the organic compound is

(1)10.5 % (2) 21%

(3) 42 % (4) 31.5 %

197. With raise in temperature of a liquid, the viscosity

(1) Increases

(2) Decreases

(3) Remains constant

(4) May increase or decrease

198. The structure of secbutyl group in an organic compound is

(1) $CH_3-CH_2-CH_2-CH_2-$

(2) $CH_3-\underset{|}{\overset{CH_3}{CH}}-CH_2-$

(3) $CH_3-\underset{|}{\overset{CH_3}{C}}-\underset{CH_3}{|}$

(4) $CH_3-\underset{|}{CH}-CH_2-CH_3$

199. A mixture of 2.3g of formic acid and 4.5g of oxalic acid is treated with conc H_2SO_4. The evolved gaseous maxture is passed through KOH pellets. Weight of remains product at STP will be

(1)1.4 g (2) 3.0 g

(3) 2.8 g (4) 4.4 g

200. $1-\text{Butene}+HBr\xrightarrow[h\nu]{H_2O_2}1-\text{Bromo butane}.$

It follows

(1) Markownikoff's rule

(2) Saytzeff's rule

(3) Anti Markownikoff's rule

(4) Hoffmann's rule

SRI CHAITANYA EDUCATIONAL INSTITUTIONS,INDIA.

A.P,TELANGANA,KARNATAKA,TAMILNADU,MAHARASHTRA,DELHI,RANCHI,CHANDIGARH

SEC : SR ELITE

DATE: 04-04-2023

NEET PART TEST – 1

BOTANY

1) **4**	2) **4**	3) **2**	4) **4**	5) **4**	6) **2**	7) **1**	8) **2**	9) **3**	10) **3**
11) **4**	12) **4**	13) **1**	14) **4**	15) **2**	16) **1**	17) **4**	18) **2**	19) **3**	20) **3**
21) **2**	22) **2**	23) **2**	24) **4**	25) **1**	26) **2**	27) **1**	28) **2**	29) **4**	30) **3**
31) **4**	32) **4**	33) **1**	34) **4**	35) **2**	36) **4**	37) **1**	38) **1**	39) **2**	40) **3**
41) **4**	42) **1**	43) **3**	44) **2**	45) **1**	46) **2**	47) **2**	48) **2**	49) **1**	50) **4**

ZOOLOGY

51) **1**	52) **3**	53) **4**	54) **1**	55) **1**	56) **1**	57) **2**	58) **4**	59) **3**	60) **3**
61) **1**	62) **1**	63) **1**	64) **2**	65) **4**	66) **1**	67) **3**	68) **3**	69) **4**	70) **2**
71) **4**	72) **3**	73) **3**	74) **1**	75) **4**	76) **4**	77) **4**	78) **3**	79) **3**	80) **2**
81) **4**	82) **4**	83) **2**	84) **4**	85) **4**	86) **2**	87) **4**	88) **4**	89) **3**	90) **4**
91) **3**	92) **1**	93) **4**	94) **3**	95) **4**	96) **2**	97) **3**	98) **3**	99) **4**	100) **3**

PHYSICS

101) **1**	102) **2**	103) **2**	104) **2**	105) **3**	106) **1**	107) **2**	108) **2**	109) **3**	110) **4**
111) **1**	112) **2**	113) **2**	114) **1**	115) **1**	116) **3**	117) **4**	118) **2**	119) **2**	120) **1**
121) **3**	122) **3**	123) **2**	124) **3**	125) **1**	126) **4**	127) **2**	128) **1**	129) **2**	130) **1**
131) **3**	132) **2**	133) **1**	134) **2**	135) **4**	136) **2**	137) **2**	138) **1**	139) **3**	140) **3**
141) **3**	142) **3**	143) **2**	144) **1**	145) **3**	146) **1**	147) **1**	148) **1**	149) **4**	150) **1**

CHEMISTRY

151) **2**	152) **1**	153) **3**	154) **4**	155) **1**	156) **4**	157) **2**	158) **1**	159) **3**	160) **4**
161) **2**	162) **4**	163) **1**	164) **3**	165) **1**	166) **4**	167) **3**	168) **1**	169) **3**	170) **1**
171) **3**	172) **3**	173) **1**	174) **4**	175) **1**	176) **2**	177) **1**	178) **3**	179) **1**	180) **3**
181) **1**	182) **1**	183) **3**	184) **3**	185) **3**	186) **3**	187) **1**	188) **1**	189) **4**	190) **3**
191) **1**	192) **2**	193) **4**	194) **3**	195) **4**	196) **2**	197) **2**	198) **4**	199) **3**	200) **3**